Week 43: Shuffling picture cards

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This view repeats, by problem, the answer key of the print guide, week-43-facilitator.pdf, whose source is editable/facilitator-src/guide.json. Change both together; check_guide_copy.py compares them.

Mathematical overview

Mathematics before the session

A fair shuffle of three distinct cards gives each of the six orders probability 1/6. Choosing the first card uniformly from three, the second uniformly from the two remaining, and the last by necessity does this: every final order has exactly one first-two choice story, of probability (1/3)(1/2).

The ticket-and-swap version is Fisher-Yates in a left-to-right form. From slots 1, 2, 3, first swap slot 1 with a uniformly chosen slot in {1,2,3}; then swap slot 2 with a uniformly chosen slot in {2,3}. The six equally likely ticket stories give six distinct orders. Swapping a slot with itself is allowed. After a position is settled, later swaps do not revisit it.

For n distinct cards, at step i choose uniformly among slots i through n and swap that slot with i. The last slot is forced. There are n! equally likely stories, and each target order determines a unique story: choose the current location of its first card, then of its second, and so on. Thus any fixed starting order produces a uniform output.

What can go wrong

The tempting rule that swaps slot 1, then 2, then 3 with a fresh choice from all three slots has $3^3 = 27$ equally likely stories. Uniformity among six outputs would require 27/6 histories per output, which is not an integer. In order ABC, ACB, BAC, BCA, CAB, CBA, the exact counts are 4, 5, 5, 5, 4, 4. Random choices at each step alone do not make the final order uniform.

Forbidding self-swaps also changes the mechanism: the printed alternative, swapping 1 with 2 or 3 and then swapping 2 with 3, produces only BCA and CAB. They can be equally likely without constituting a fair shuffle of all six orders. This rule is Sattolo's algorithm for three cards. For n cards it makes each of the $(n-1)!$ orders that form a single cycle equally likely, so no card ever stays in its starting slot. Related weeks: 3 (Shuffle machines) for the cycles of fixed shuffles, 63 (Cards away from home) for rows with no card at home, and 23 (Sorting machines), 42 (Fair results from a bag) and 44 (The bag that copies) for weighing equally likely histories.

One fixed swap history is a bijection of starting orders, so it cannot merge two different starts. This does not conflict with the uniform distribution from every start when the history itself is random. Distinguish "the same history gives the same answer" from "the random answers have the same distribution."

Assumptions and discoveries by band

All eligible tickets must have equal conditional chances at each fresh draw; reset the ticket sets between full shuffles. Cards must be distinct and equally selectable. A small sample need not show each order equally often. **K-1:** build all six orders, follow the two swaps, and investigate missing choices. **Grades 2-3:** match histories to outputs and contrast a six-story rule with a 27-story rule. **Grades 4-5:** prove the bijection, find algorithm failures, and design four- and five-card versions.

Problems and routes

Problem	K–1	2–3	4–5	route
Every row of three cards	P1			first
Every row the chooser can make	P2	P1	P1	first
The six two-ticket stories	P3	P2	P2	first
A different starting row		P3	P3	2-3 return; 4-5 first
A ticket goes missing	P4			return
One story, two starting rows	P5			return
Must six shuffles show all six rows?		P4		return
Different stories, same row		P5		return
Twenty-seven stories, six rows		P6	P4	return
Forbid swapping a slot with itself			P5	return
Design a fair four-card shuffle			P6	return
Design a fair five-card shuffle			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P2; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P5; Grades 2–3 P3–P6; Grades 4–5 P4–P7.

Key, one entry per problem

Every row of three cards

K–1 P1 · first route

Idea. Three distinct cards make $3 \times 2 = 6$ rows: a choice of first card, then of second.

The six rows are ABC, ACB, BAC, BCA, CAB, CBA: three choices of first card and two orders of the remaining cards.

Every row the chooser can make

K–1 P2 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Each order has exactly one first-two draw story, of probability (1/3)(1/2) = 1/6, so the chooser is fair.

K–1: Each displayed card is the card drawn first. If **A** is first, the rows are ABC and ACB; if **B** is first, BAC and BCA; if **C** is first, CAB and CBA. Two different first-two draws cannot make the same row because that row displays both draws in its first two positions.

Grades 2–3: ABC, ACB, BAC, BCA, CAB, CBA each has one first-two draw story: AB, AC, BA, BC, CA, CB respectively. Every story has probability $1/3 \times 1/2 = 1/6$. Therefore none is favored. This proof is about all possible choices, not six observed runs.

Grades 4–5: Direct-choice stories AB, AC, BA, BC, CA, CB produce ABC, ACB, BAC, BCA, CAB, CBA. Each story has probability 1/6.

The six two-ticket stories

K–1 P3 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. Left-to-right Fisher-Yates: the six equally likely ticket stories give six different rows, one per order.

K–1: Write a ticket story as (first ticket, second ticket), not as card names. The complete correspondence from ABC is:

Ticket story	Final row	Ticket story	Final row
(1,2)	ABC	(1,3)	ACB
(2,2)	BAC	(2,3)	BCA
(3,2)	CBA	(3,3)	CAB

No final row comes from two different stories. First ticket 1, 2, or 3 puts A, B, or C first, and the second ticket chooses between the two remaining orders.

Grades 2–3: In the printed order, the mappings are (1,2) → ABC; (2,2) → BAC; (3,2) → CBA; (1,3) → ACB; (2,3) → BCA; (3,3) → CAB. The six stories are equally likely, and every row appears once, so this rule is fair.

Grades 4–5: For the two-swap rule from ABC, the complete list is (1,2) → ABC, (1,3) → ACB, (2,2) → BAC, (2,3) → BCA, (3,2) → CBA, (3,3) → CAB. Again each equally likely story has a different output, proving uniformity.

A different starting row

Grades 2–3 P3 · Grades 4–5 P3 · 2-3 return; 4-5 first route · wording differs by band

Idea. Every target order still has exactly one story from any start: choose where its first card is, then its second.

Grades 2–3: Starting with BAC changes the target-to-ticket correspondence, but all six rows remain possible and equally likely.

Target	Ticket story	Target	Ticket story
ABC	(2,2)	ACB	(2,3)
BAC	(1,2)	BCA	(1,3)
CAB	(3,2)	CBA	(3,3)

Grades 4–5: Every order still has chance 1/6 from CBA. Relabel A and C throughout the original experiment: its start becomes CBA, the allowed ticket histories and their weights do not change, and the output labels are permuted bijectively. Alternatively, for any target choose its first card from the three current slots, then its second from the two remaining slots; this determines a unique story regardless of the starting order.

A ticket goes missing

K–1 P4 · return route

Idea. Removing first ticket 1, 2 or 3 means A, B or C can never be first, so exactly two rows become impossible.

Remove first ticket **1**: ABC and ACB become impossible. Remove **2**: BAC and BCA become impossible. Remove **3**: CAB and CBA become impossible. Each removal prevents one particular card from ever being first. The second set remains {2,3}.

One story, two starting rows

K–1 P5 · return route

Idea. A fixed swap story can be undone in reverse order, so it never sends two different starts to the same row.

No. A swap is undone by repeating it. If the finishes matched, undoing the shared swaps would recover equal starts, contradicting ABC versus CBA. Here are all six comparisons.

Story	Finish from ABC	Finish from CBA
(1,2)	ABC	CBA
(1,3)	ACB	CAB
(2,2)	BAC	BCA
(2,3)	BCA	BAC
(3,2)	CBA	ABC
(3,3)	CAB	ACB

Must six shuffles show all six rows?

Grades 2–3 P4 · return route

Idea. Equal chances do not make a short run representative: six fair shuffles show all six rows with probability $6!/6^6 = 5/324$, about 1.5%.

No. Six shuffles have $6^6 = 46{,}656$ equally likely lists of ticket stories, and only $6! = 720$ of them show all six rows once, about 1 run in 65. Six repeats of (1,2), giving ABC six times, is one legal run. Uniform probability does not force a perfectly representative batch of six results.

Different stories, same row

Grades 2–3 P5 · return route

Idea. Under the all-slots rule, ABC has four stories: (1,2,3), (1,3,2), (2,1,3) and (3,2,1); a swap and its repeat cancel.

Under the new three-swap rule, four different histories produce ABC: (1,2,3), (1,3,2), (2,1,3), and (3,2,1). For example, the first uses three self-swaps; the second swaps the last two twice; the third swaps the first two twice; the fourth swaps the first and last twice. These are exact collisions in the rule, not repeated observations of a single story. Other valid collisions are acceptable.

Twenty-seven stories, six rows

Grades 2–3 P6 · Grades 4–5 P4 · return route · wording differs by band

Idea. 27 equally likely stories cannot split equally among 6 rows, since 6 does not divide 27; the counts are 4, 5, 5, 5, 4, 4.

Grades 2–3: No. There are $3 \times 3 \times 3 = 27$ equally likely histories. If all six rows were equally likely, each would need the same whole number of histories. But six equal integer counts cannot sum to 27. The exact distribution below is for the facilitator; enumeration is not required of children.

Final order	ABC	ACB	BAC	BCA	CAB	CBA
History count	4	5	5	5	4	4

The fact that there are collisions alone does not prove unfairness: all outputs might have had the same number of histories. The unequal allocation forced by 27 versus 6 supplies the proof.

Grades 4–5: No. The 27 equally likely histories cannot be divided into six equal-sized output classes. Exact counts for ABC, ACB, BAC, BCA, CAB, CBA are 4, 5, 5, 5, 4, 4. The argument relies on equiprobable histories; merely knowing a total number of possible histories would not suffice if they had unequal weights.

Forbid swapping a slot with itself

Grades 4–5 P5 · return route

Idea. Without self-swaps only BCA and CAB occur: two orders at 1/2 each, not a fair shuffle of six.

First swap 1 with 2: ABC $\to$ BAC $\to$ BCA after swapping 2 with 3. First swap 1 with 3: ABC $\to$ CBA $\to$ CAB. Thus only BCA and CAB occur. If the initial two choices are equally likely, these have probability 1/2 each, but four orders have probability zero. It is not a fair shuffle of all six orders.

Design a fair four-card shuffle

Grades 4–5 P6 · return route

Idea. Ticket sets {1,2,3,4}, {2,3,4}, {3,4} give $4 \times 3 \times 2 = 24$ equally likely stories, exactly one per order.

For ABCD, swap slot 1 with a uniform ticket from {1,2,3,4}; swap slot 2 with a uniform ticket from {2,3,4}; swap slot 3 with a uniform ticket from {3,4}. Leave slot 4. Permit self-swaps. At each stage empty the cup and use exactly the indicated ticket set; for a new full shuffle reset the starting row and all ticket sets. There are $4 \times 3 \times 2 = 24$ equally likely stories.

A target order uniquely specifies the ticket at each step: locate the target first card, swap it into slot 1, then locate the target second card in the suffix, then the third. Later steps cannot disturb a settled slot. Consequently each of the 24 orders has exactly one story, of probability 1/24.

Other designs pass too. Keeping the swaps of slot 1, then 2, then 3, exactly six choices of ticket sets give one story per order, one for each order of the set sizes 4, 3, 2; for example {1,4}, {1,2,4}, {1,2,3,4}.

Design a fair five-card shuffle

Grades 4–5 P7 · return route

Idea. Ticket sets {1,2,3,4,5}, {2,3,4,5}, {3,4,5}, {4,5} give 120 equally likely stories, one per order.

Use sets {1,2,3,4,5}, {2,3,4,5}, {3,4,5}, {4,5} for swaps of slots 1, 2, 3, 4 respectively, then leave slot 5. Reset the cup to the prescribed set before each choice and reset the full procedure for each new shuffle. With uniform fresh choices, $5 \times 4 \times 3 \times 2 = 120$ histories have equal probability. The same unique-target construction proves one story per order, hence probability 1/120 for each.

Optional hints and facilitation checks

For completeness: “Fix the first card; what can happen to the last two?” For swaps: “Which slot contains the card you want first?” For the wrong rule: “How many equally likely stories would each row need?” For identical histories from different starts: “Can you undo each swap in reverse order?” To decide fairness: “Use every possible choice story and its final row.” A short run of random shuffles cannot settle the question.

Source and verification

Robert Sedgewick and Kevin Wayne, Princeton *Algorithms*, Knuth.java (algs4.cs.princeton.edu/11model/Knuth.java.html), especially shuffleAlternate and its uniform-independent random-source assumption. Their Elementary Sorts lecture, slide 71 (PDF p. 18), identifies the wrong-range bug and the valid suffix-range variant. The three-card enumerations and child tasks here are independently checked adaptations.